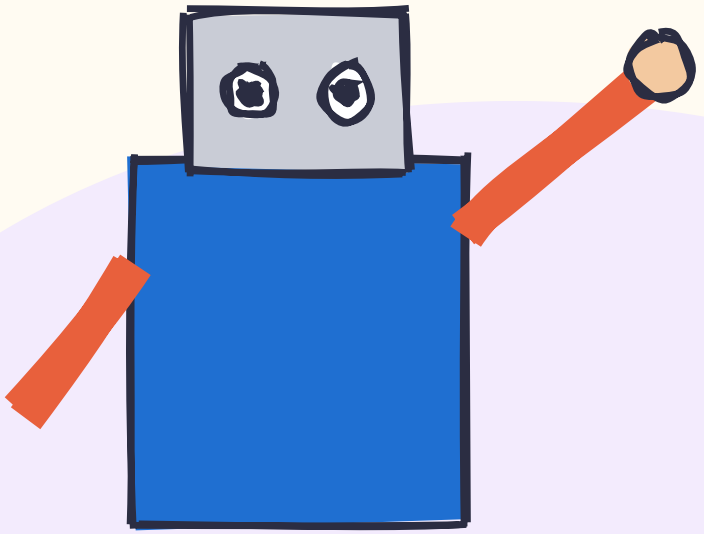
Little Explorers · Class 9

Waving Robot Arm

Build a cardboard robot that waves hello!

Servo motor | Angles | Robot design



Did you know?
Some airports and hospitals have robots that greet visitors and show them the way!

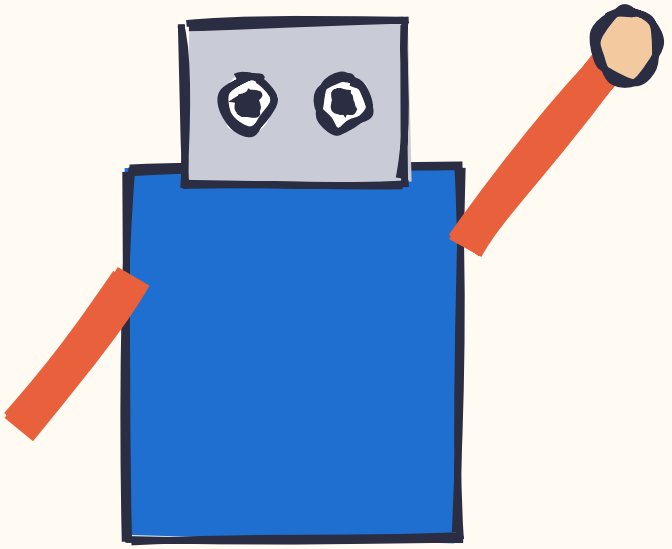
LEARN

Let's wonder...

How does a robot know how far to move?

Hello, robot!

When you wave, your brain tells your arm muscles exactly how far to go. A robot has no muscles. What moves its arm, and how does it know where to stop?



Strings?

A spring?

A motor and code!

Today: we give a cardboard robot a **motor arm** and code it to wave!

LEARN

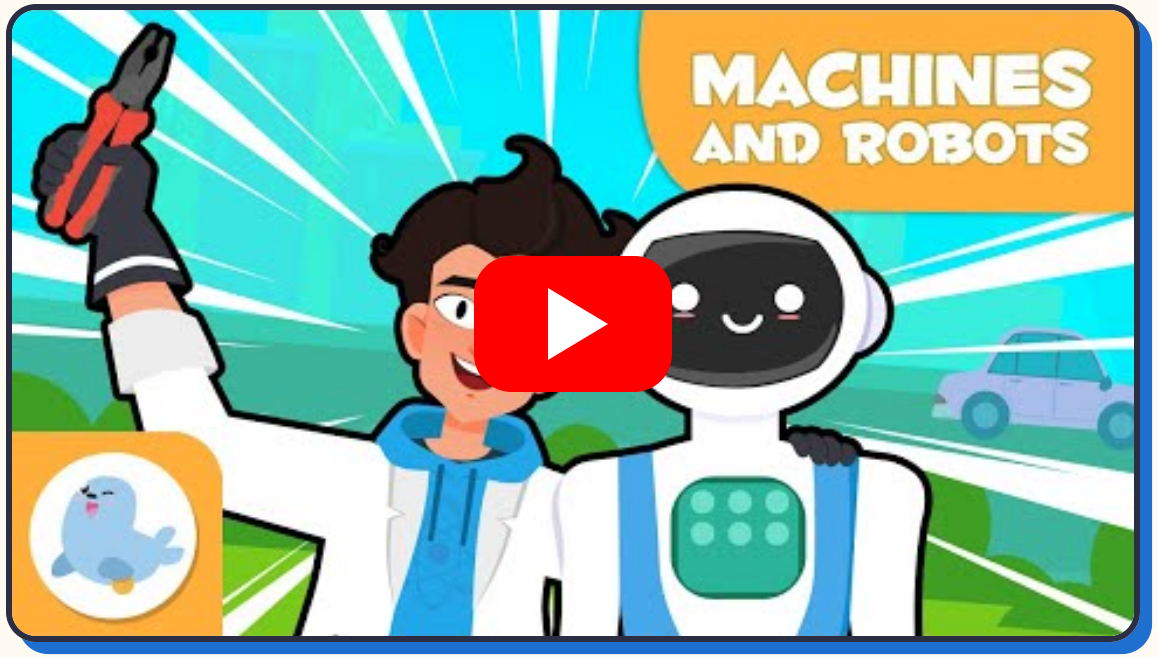
Learning objectives

Today I can...

1	**Servo:** say that a **servo motor** turns to an exact position	**Proof:** show my servo turning to two places
2	**Angles:** choose two **angles** and repeat them to make a wave	**Proof:** my robot waves hello
3	**Design:** design a **robot character** with a name	**Proof:** my robot has a name and a face

Objective 1 · Servo · video

Video time!



Machines and Robots for Kids

Smile and Learn · about 4 minutes · youtu.be/PMvrvaN8YRM

Watch for...

1. What is the difference between a **machine** and a **robot**?
2. Name a robot that **moves its arms**.
3. How do robots know what to do?

Scan to watch

After the video: Be a robot! Your partner gives you 3 instructions. Follow them exactly, like a robot.

Objective 1 · Servo

A servo turns to an exact spot

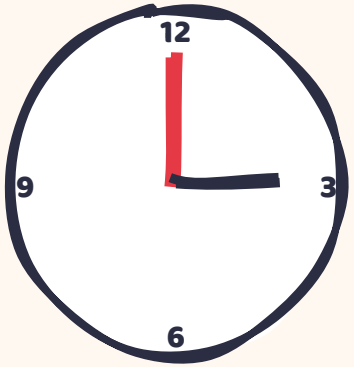
The idea

A normal motor just spins round and round. A **servo motor** is special: it turns to an **exact position** and **stops** there.

We tell it a number, like **0**, **90** or **180**, and it turns to that spot.

Servos everywhere
Servos are inside remote-control cars, robot arms and even some toys!

Example: a clock hand

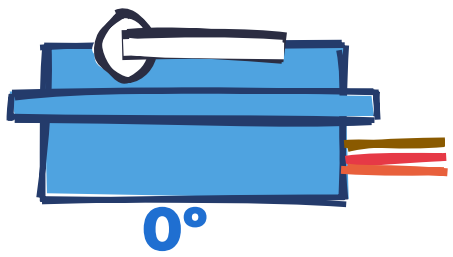


A clock hand points to **exact** numbers: 12, 3, 6, 9. A servo arm is like a clock hand that we can move to any number we choose!

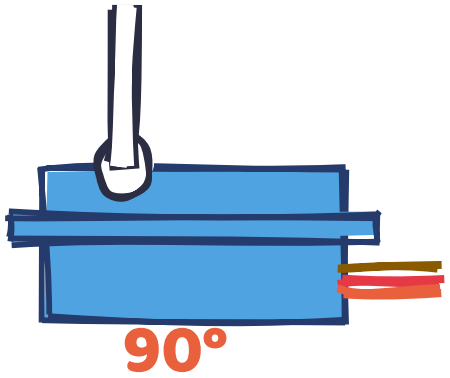
LEARN

Objective 2 · Angles

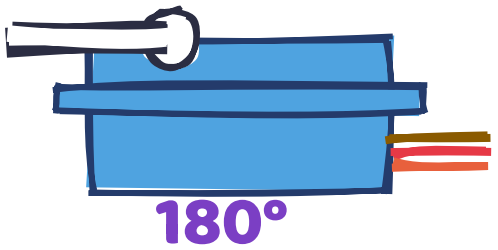
Angles: how far to turn



arm flat



arm straight up



arm flat the other way

A wave

To wave, the arm goes to **45°**, waits, then to **135°**, waits… and repeats!

Arm game

Everyone show 0°, 90° and 180° with your arm!

Real world

Robots that move arms



Car factories
Robot arms build and paint cars



Hospitals
Robot arms help doctors with tiny, careful work



Space
A robot arm on the space station catches spaceships



Welcome robots
Wave and talk to visitors

? If you had a robot arm at home, what job would you give it?

Gentle and strong
Robot arms can lift a car or pick up an egg without breaking it!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Robot Says
2. Make your robot
3. Snap the picture blocks
4. Fast wave or slow wave?
5. Make your robot dance!
6. Robot arm challenges

Class game

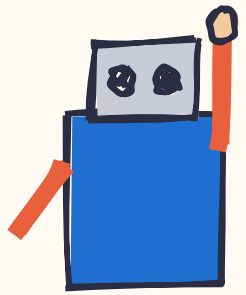
Robot Says

How to play

1. Like 'Simon Says', but with angles!
2. Teacher says **"Robot says 90!"**: arms straight up.
3. **"Robot says 0!"**: arms flat out.
4. No 'Robot says'? Do not move!
5. Last robot standing wins.

Then discuss

Was it easy to stop at **exactly** the right place? Servos are very good at this!



Big idea

Robots move exactly where the code tells them, every single time.

MAKE

Objective 3 · Build step 1 of 2

Make your robot

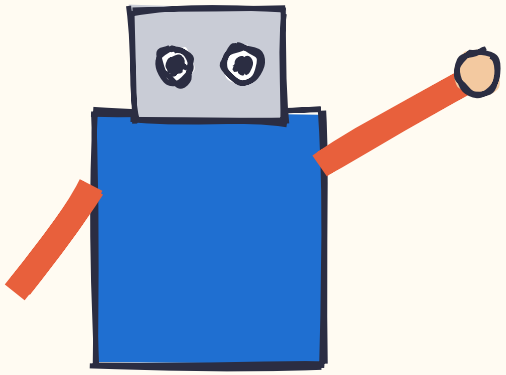
a Draw a robot face and body on the cardboard. Add **googly eyes**!

b Cut out one **arm** from cardboard.

c Stick the arm onto the white **servo horn** with tape.

d Stick the servo to the back of the robot so the arm sticks out the side.

e Give your robot a **name**!



Check: The arm is firmly taped to the servo and the robot stands up.

Ask a grown-up to help with the scissors.

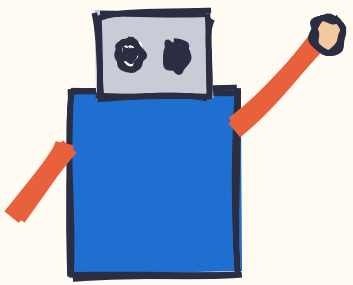
Time to build!

Snap the picture blocks

```
when button (pin 2) is pressed
repeat 3 times
    set servo (pin 9) angle to 45
    wait 0.5 seconds
    set servo (pin 9) angle to 135
    wait 0.5 seconds
set servo angle to 90 (rest)
```

Read it like a story

When I press the button:
arm to 45, wait, arm to 135, wait.
Repeat 3 times = 3 waves!
Then rest at 90.



Repeat
Repeat means do the same thing again. It saves us snapping lots of blocks!

MAKE

Experiment!

Fast wave or slow wave?

Change the wait to…	I think the wave is…	What happened
wait 1 second	slow / fast?	_____
wait 0.5 seconds	slow / fast?	_____
wait 0.2 seconds	slow / fast?	_____

Talk about it

Which wave looks **friendly**? Which looks **excited**? Which looks **sleepy**?

Gentle!

Very fast waves can shake the robot apart. Hold the cardboard steady.

MAKE

Objective 2 · Angles · be creative

Make your robot dance!

Move	Angle	Wait
1	_____°	_____ s
2	_____°	_____ s
3	_____°	_____ s
4	_____°	_____ s

Plan your dance

Choose 4 angles and 4 wait times. Write them in the table. Then snap the blocks in the same order!

Robot show

Each robot does its dance for the class. Clap along!

MAKE

Level up!

Robot arm challenges

Choose your challenge



Hello tune: play a beep when the robot waves.



Light-up eyes: add an LED that turns on while it waves.



Two arms: can a friend's servo make the other arm wave too?



High five: slowly raise the arm to 180 for a high five.

Think like an engineer

What jobs could a waving robot do at our school?

Boss level

Make the robot wave only when someone is near (use a sensor)!

Objective check

Can I really do it? Prove it.

01 What is special about a servo motor?

02 Show your robot waving. Which two angles did you use?

03 Tell us your robot's name and its job.

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Robot parade: each group shows their robot and says one angle they used.

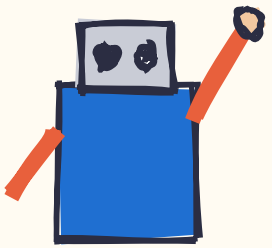
Show what you know

What did we learn?

- [x] A servo motor turns to an exact angle
- [x] 0°, 90° and 180° are different positions
- [x] Two angles + repeat = a wave
- [x] Robots move exactly as the code says

At home

Show your family a wave at 0°, 90° and 180° with your arm.



Exit ticket: which number makes the servo point straight up?

Next class: build an Automatic Railway Crossing!